Matched COBA and PING on a full-development SHD split

exp068 · 19 July 2026 · Draft

Abstract

This one-seed exploratory experiment asks whether PING's modest `exp066` point advantage becomes clearer with substantially more Spiking Heidelberg Digits (SHD) development data and leakage-resistant model selection. COBA and PING train on the same 7340 utterances, select checkpoints using the same validation split of 816 utterances, and are evaluated once on all 2264 official-test utterances after both checkpoints freeze. PING minus COBA official-test accuracy is 10.64 percentage points, which meets the registered interpretation band **promising PING accuracy claim**. This single seed identifies an exploratory direction; it does not estimate an expected architecture effect.

Methods

Data separation prevents test-driven selection

The official SHD training split of 8156 utterances is partitioned at seed 42 into 7340 training and 816 validation utterances. Stratification is joint by speaker and class. Both architectures use the same index hashes. The complete official test is unavailable during training and checkpoint selection. Only after both checkpoint identities are frozen is each checkpoint evaluated once on the complete official test. Speakers 4 and 5, absent from official training, define the registered unseen-speaker diagnostic.

The two cells differ only in the inhibitory recipe

Both networks have 256 excitatory and 64 inhibitory cells, fixed Dale-constrained recurrence, input-weight mean 0.9, 95% input sparsity, a membrane-mean readout scaled by 225, Adam learning rate 0.0004, batch size 32, a 1 ms timestep, and 1000 ms utterance duration. COBA disables the inhibitory loop and uses no voltage-gradient dampening; PING enables the loop and uses dampening 1000. Neither cell uses a firing-rate regulariser.

Each cell trains for 40 epochs. The selected checkpoint maximises validation accuracy, breaking ties with lower validation cross-entropy and then the earlier epoch. The experiment-side selector preserves the required epoch state without modifying the SNN tool.

Results

Validation selects checkpoints without official-test feedback

COBA selects epoch 25 at 40.81% validation accuracy; PING selects epoch 40 at 43.5%.

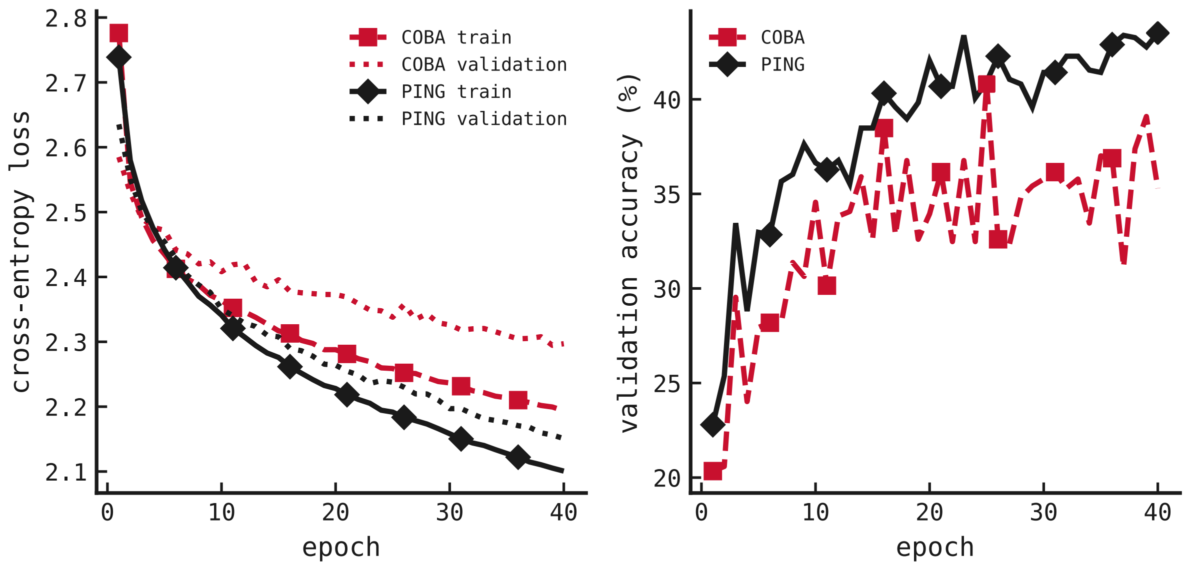

Figure 1: Training and validation learning curves under the locked split. Red dashed traces and square markers denote COBA; black solid traces and diamond markers denote PING. The marked epoch for each cell is selected solely by validation accuracy, validation cross-entropy, and the registered earlier-epoch tie-break.

The official test determines the exploratory accuracy verdict

COBA obtains 35.07% overall official-test accuracy and PING obtains 45.72%. Their registered difference is 10.64 percentage points. The macro-class, seen-speaker, and unseen-speaker panels are diagnostics rather than additional primary outcomes.

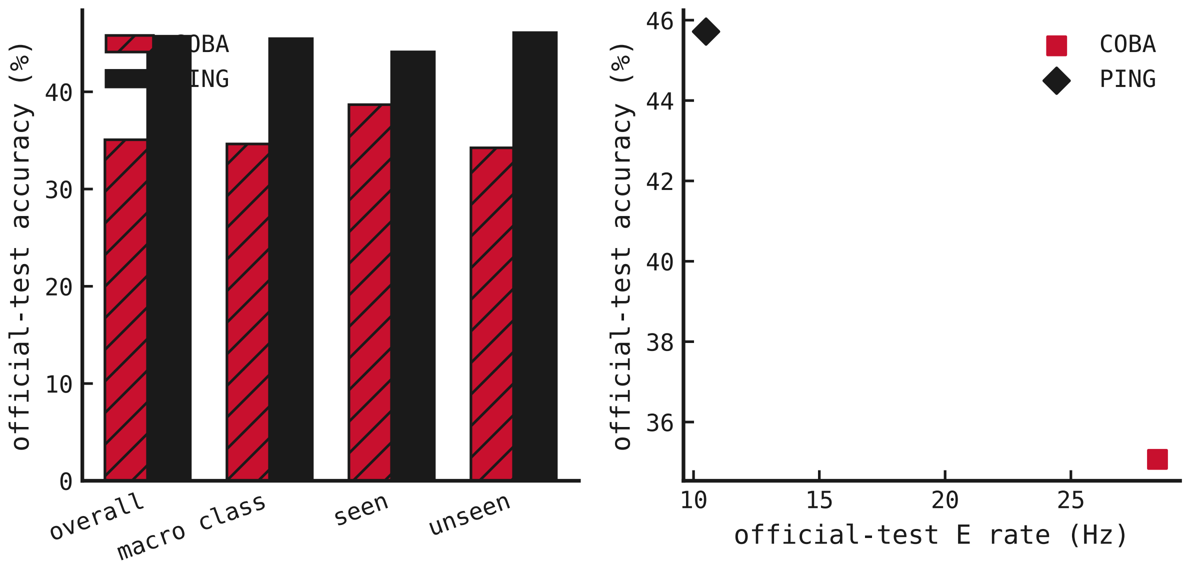

Figure 2: Official-test results after both validation checkpoints were frozen. Left: overall accuracy, macro class accuracy, and accuracy for speakers seen versus unseen during development training. Right: the same overall accuracy plotted directly against official-test excitatory firing rate; no post-hoc composite score is used.

PING uses less excitatory activity under the matched recipe
On the official test, COBA's excitatory population fires at 28.44 Hz. PING fires at 10.5 Hz excitatory and 47.99 Hz inhibitory. Validation activity is shown across all epochs so activity drift is visible rather than reduced to a single selected point.

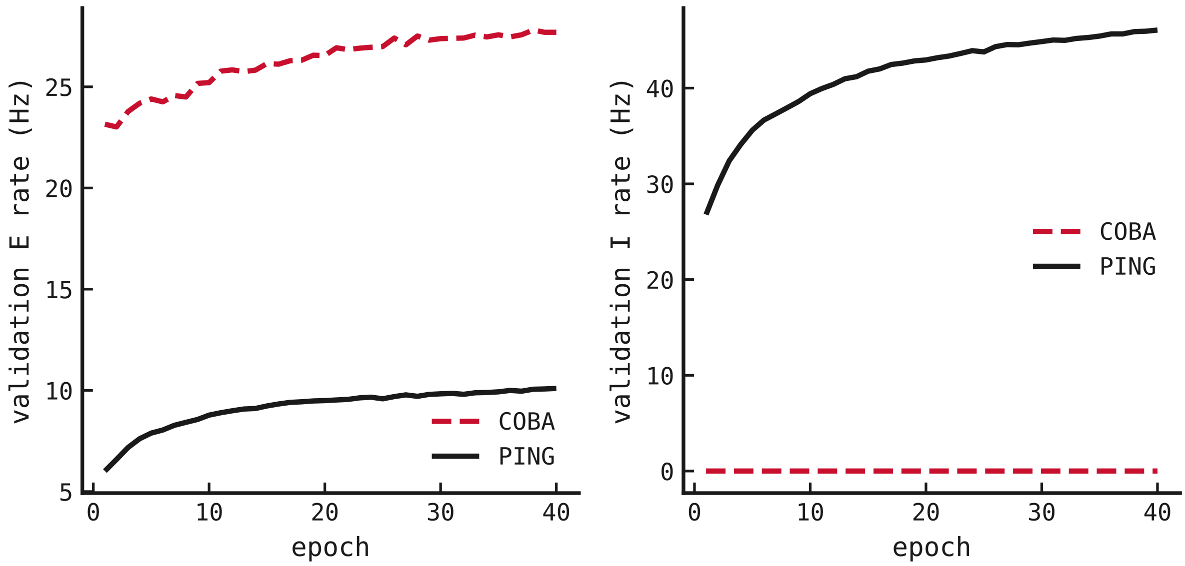

Figure 3: Population firing rates on the validation split during training. COBA's disabled inhibitory loop stays at zero. PING sustains active E and I populations; the accuracy comparison remains matched on every non-architectural setting.

Fixed utterances expose matched population dynamics

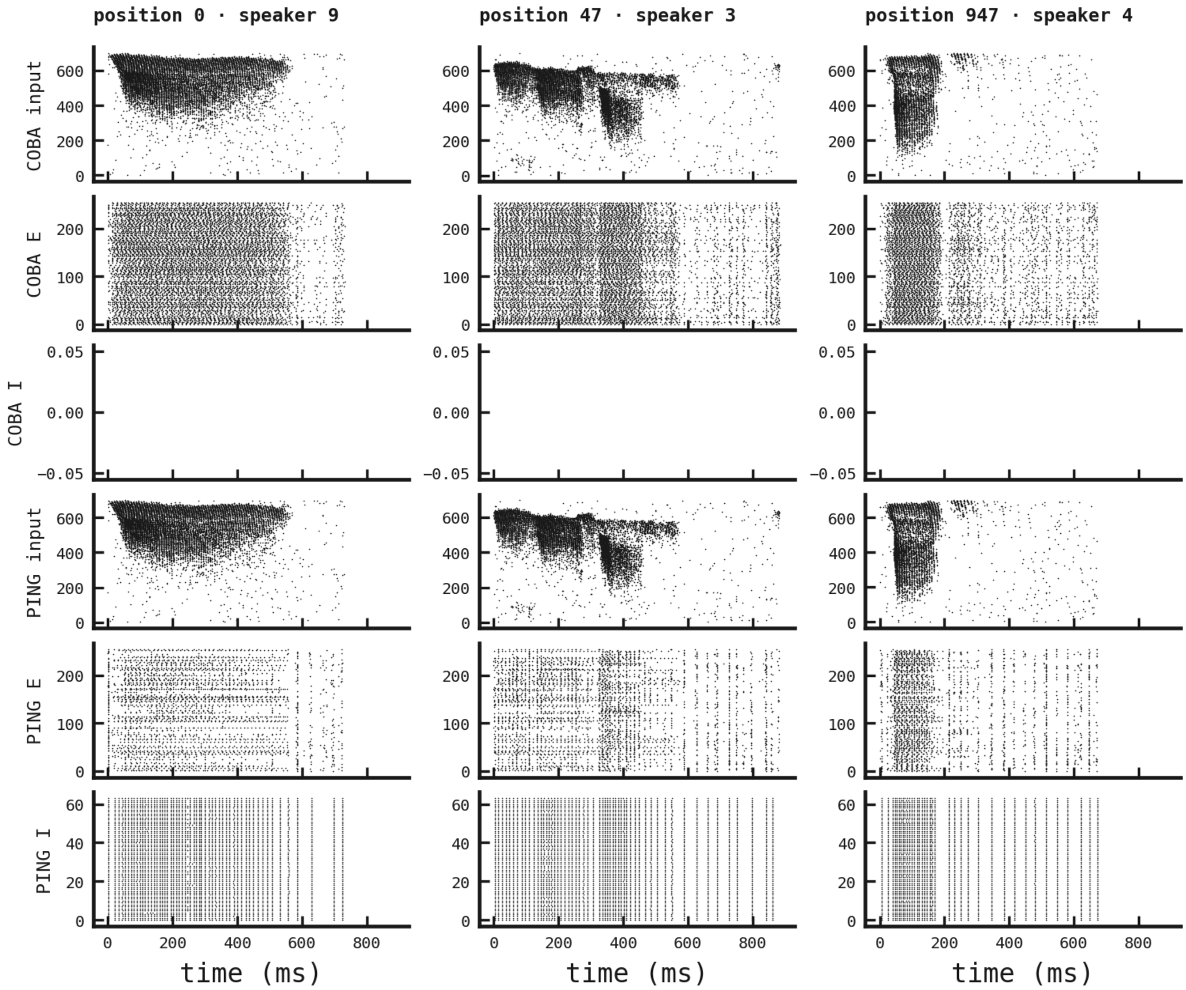

Figure 4: Matched spike rasters for official-test positions 0, 47, 947, fixed before predictions were inspected. Each column uses byte-identical binned cochlear input in COBA and PING. Rows show input, excitatory, and inhibitory spikes on the same time axis and fixed population order.

Verdict and limits

The registered exploratory interpretation is **promising PING accuracy claim**: PING minus COBA accuracy is 10.64 percentage points against the predeclared +2-point promising-signal threshold. This conclusion applies to one seed and one matched recipe. It cannot support uncertainty intervals or a population-level architecture claim. A later confirmatory mandate is warranted only if this exploratory result supplies a claim worth defending.

Both cells used identical partition hashes, official-test evaluation occurred after the paired checkpoint freeze, and neither cell recorded a skipped update or non-finite forward batch. Total RunPod spend was 2.441 USD, with 0 active pods after collection.